Gears

Concepts

- Gears are mechanical devices that transfer rotation from one power shaft to another via interlocking teeth, commonly used to exchange a decrease in angular speed ω for an increase in torque T. The **efficiency** of gears is given by $\eta = P_{out}/P_{in} = (T_{out}\omega_{out})/(T_{in}\omega_{in})$.
- In this class, we will discuss the design of spur gears and helical gears. **Helical gears** are characterized by two mating cylinders having parallel rotation axes, with teeth cut on the outside of the cylinders that mesh together to transfer torque and rotation. The angle that teeth are cut relative to the rotation axis is called the **helix angle** ψ. A **spur gear** is a helical gear with $\psi = 0$.
- Since the teeth (and thus the outer diameters) of two gears are designed to overlap, we define a gear's **pitch circle** upon which to base calculations. Two gears will mate correctly when their pitch circles are tangent.
- Along with ψ, a helical gear is chiefly characterized by a **pitch diameter** d (the diameter of its pitch circle), a transverse pressure angle ϕ (the angle at which two mating gear teeth make contact relative to the line connecting their centers), and an integer **number of teeth** N.
- For helical gears, sometimes a **normal pressure angle** ϕ_n is provided instead of ϕ. They are related by $\cos\psi = \frac{\tan\phi_n}{\tan\phi}$.
- Two mating gears must have equal **circular pitch** p (the distance along the pitch circle between corresponding points from two adjacent gear teeth) which is defined to be $\pi d/N$. The **diametral pitch** P must also be equal, and is defined as N/d.
- The smaller of two mating gears is typically called the **pinion**, while the larger is typically the **gear**. Given a pinion with N_P teeth and a gear with N_G teeth, their **gear ratio** is defined as $m = N_G/N_P \geq 1$. The angular speeds of a pinion and gear are related by $N_P\omega_P = N_G\omega_G$.
- **Interference** occurs when two gears make contact with corners of teeth rather than via normal surface contact, leading to wear and poor performance. To avoid interference, the number of teeth on the pinion must be at least:

$$N_P \geq \frac{2k\cos\psi\left(m + \sqrt{m^2 + (1+2m)\sin^2\phi}\right)}{(1+2m)\sin^2\phi}$$

  where $k = 1$ for **full-depth teeth** (default) and $k = 0.8$ for **stub teeth**; and the number of teeth on the gear can be at most:

$$N_G \leq \frac{N_P^2\sin^2\phi - 4k^2\cos^2\psi}{4k\cos\psi - 2N_P\sin^2\phi}$$

- A **rack** is a straight gear that can be thought of as a subset of a gear with infinite pitch diameter. A rack and pinion system effectively has gear ratio $m \to \infty$, so the pinion must satisfy:

$$N_P \geq \lim_{m\to\infty} \frac{2k\cos\psi\left(m + \sqrt{m^2 + (1+2m)\sin^2\phi}\right)}{(1+2m)\sin^2\phi} = \frac{2k\cos\psi}{\sin^2\phi}$$

Practice Problems

1) A 15-tooth spur pinion with 10 mm circular pitch rotates at 1600 rev/min to drive a 60-tooth gear. Determine the speed of the driven gear and their theoretical center-to-center distance.

2) Determine the smallest helical pinion with normal pressure angle 20° and helix angle 30° that will run with a rack without interference.

3) Determine the maximum gear ratio that can be achevied that avoids interference when combining a 25°-pressure angle 8-tooth stub pinion with a gear.

4) Determine the minimum pressure angle for a 9-tooth spur pinion to mesh with a rack while avoiding interference.